MATH 128A syllabus Fall 2025

Note: The information presented in this document is subject to change (with notification).
Last change: Sept 2, 2025.(added quiz weeks)

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Lectures: Tu/Th 9.40-10.59am, Haas Faculty Wing F295

Office Hours Franziska: 3-4pm Tuesdays, and 11am-12pm Thursdays at my office 805 Evans Hall

Office Hours GSIs:

Tom: 2-3.30pm Thursdays in 1041 Evans

Vinh-Kha: 2-3pm and 4-6pm Tuesdays in 1039 Evans

Shilun: TBA

Yuchen: 1-2.30pm Monday and 2-3.30pm Friday in 747 Evans

The **office hour schedule** can also be found at this google sheet: https://docs.google.com/spreadsheets/d/1wTlu-prrF1DU7-TNAoC6OCs_rIcUmvfIIGCnV6M_56c/edit?usp=sharing

Prerequisites: MATH 53 and 54, or equivalent. Basic MATLAB programming skills (e.g. MATH 98, MATH 124, CS61A, Engin 7 or equivalent). If you don't have experience with MATLAB, consider taking the 3 week course MATH 98 at the beginning of the semester. Here is the course website for this semester. More information about MATLAB can be found below.

Course description: Programming for numerical calculations, round-off error, approximation and interpolation, numerical quadrature, matrix computations, and numerical solutions of ordinary differential equations.

Topics:

- Error analysis, roundoff error and computer arithmetic, algorithms and convergence (Chapter 1)
- Nonlinear equations: bisection, Newton and secant methods (Chapter 2)
- Polynomial interpolation and approximation (Chapter 3)
- Numerical differentiation and integration (Chapter 4)

- Initial value problems for ordinary differential equations (Chapter 5)
- Matrix computations: linear systems, matrix factorizations, norms (Chapter 6, 3.5, 7.1)

Course resources:

It is your responsibility to check these resources regularly.

bcourses (canvas): Official course website, contains links to Ed Discussion, Gradescope, syllabus etc. Please e-mail me (fweber@berkeley.edu) if you do not have access to the bCourses page by the end of the second week of classes. All essential official announcements will be made on bCourses (not on Ed), in particular policy messages and practical details. Note that we are not using the grading features of bCourses, so the percentages reported are not accurate.

Gradescope: This is where you will upload completed assignments and can access your graded assignments, quizzes and exams. If you would like to request a regrade for a problem, you have to make this request in person in office hours within two weeks following the release of the corresponding graded exam/quiz/programming assignment. There is a link in bcourses to access Gradescope.

Ed Discussion (formerly piazza): Will be used for discussions about the material. It is primarily intended for discussions between the students, but the instructors will also be there to help answer questions (although perhaps not as frequently as other students). If you need more direct communication with the instructors, please bring it up during lectures, or discuss it in office hours. You can configure your settings to get email notifications of new postings. There is a link on bcourses to access Ed Discussion.

Textbook:
R. L. Burden and J. D. Faires, Numerical Analysis, 10th edition, Cengage Learning, 2015.
ISBN-13: 978-1305253667
Note: The 9th edition is also supported.

Additional reading material will be suggested in class or made available on the course website.

Matlab resources:

- Textbook by Cleve Moler (Matlab founder): Pdfs of the chapters are available for free at: https://www.mathworks.com/moler/chapters.html . The first chapter is an introduction to Matlab.
- Tutorials and documentation on the mathworks website: https://matlabacademy.mathworks.com/details/matlab-onramp/gettingstarted and https://www.mathworks.com/help/matlab/getting-started-with-matlab.html
- Within Matlab, use the command 'help' or go to the 'Help' button in the menu.

- Matlab cheat sheet: https://www.mathworks.com/matlabcentral/fileexchange/47533-cheatsheet-pdf
- Check out the resources on the MATH98 website: https://math.berkeley.edu/~tom_schang/my-website/course/math98_fa25/
- More tutorials and installation instructions from the UC Berkeley Software website: https://software.berkeley.edu/matlab%C2%AE
- If you don't want to install Matlab, you can also use Matlab online: https://www.mathworks.com/products/matlab-online.html (you need to create an account with your university email) or Octave Online https://octave-online.net/ (may have slight differences)

Ways to run matlab:

MATLAB is available in the computer lab B3A Evans.
The University has a site license you can use to install it on your own computer. You can download MATLAB here and obtain a student license here.

Schedule

A log of what was treated in each class meeting will be available at this link. (https://docs.google.com/spreadsheets/d/1f1Bqrb_pzaImp03hoJiSnTj6XMJiwjIPxUoXtU3h6cM/edit?usp=sharing)

Homework and programming projects

- The weekly homework assignments will involve theoretical and MATLAB coding exercises. The homework will be graded by completion/effort. You can gain 0-2 points per homework up to a maximum score of 22 points (you can think of this roughly as the two lowest homework scores being dropped).
- There will be 3 programming assignments for this course. They will be posted about 2-3 weeks before the due date. They should be submitted on gradescope with MATLAB code as well as report and any program outputs requested on the assignment sheet. They will be graded by completion/effort and the lowest score of the three will be dropped.
- The dates for the programming assignments are:
  - Assignment 1: Due 9/25/2025
  - Assignment 2: Due 10/23/2025
  - Assignment 3: Due 12/04/2025
- Late submissions for homeworks or programming assignments will not be accepted.
- **AI and collaboration policy**: In this course, your goal is to master the material, not just to produce correct answers. For the homework and the programming assignments, you are encouraged to learn from any available resource, including collaborating with classmates, consulting online materials (e.g., Stack Overflow), and

using AI assistants (e.g., Gemini, ChatGPT). These are powerful tools for brainstorming, debugging, and understanding complex concepts. However, the work you submit must be your own. You are responsible for understanding and being able to explain every line of your solution/code from scratch.
Think of it this way: discussing mathematical concepts and algorithms with a friend is great, and asking an AI to explain an error is smart. However, simply copying a solution from any source without understanding it is academic dishonesty. Be aware that you will be tested on these concepts in the quizzes and the exams, where these resources and collaborators will not be available. The homework is your opportunity to practice and genuinely learn the material, so make sure you are the one doing the learning. 'Paper and pencil programming' problems may be asked on quizzes and exams, so solving the assignments is great preparation for quizzes and exams!

Quizzes:

- Quizzes will take place in section biweekly. They are closed book.
- Missed quizzes can't be made up.
- The lowest quiz score will be dropped.
- The quizzes will be in the following weeks (c.f. Log file)
  - Week 3
  - Week 5
  - Week 7
  - Week 9
  - Week 11
  - Week 15
- For those who take the quizzes at the proctoring center, they should be scheduled on the Monday of the corresponding week (you need to schedule them by yourself)

Exams:

- There will be two midterm exams administered throughout the semester. They will take place in class during regular class hours. Please make sure you come to class on time to have all the allotted time to solve them. The midterms are closed book.
- The final exam will take place during the regular final examination period of the university. It is currently scheduled for 12/16/2025 3-6pm, location TBA.
- Regarding academic integrity violations, please see the section on academic integrity following later in this syllabus. Standard university procedures will be followed.
- The midterms will take place on
  - Midterm 1: 9/30/2025 (in class, 9.40-11am)
  - Midterm 2: 11/18/2025 (in class, 9.40-11am)

- Your lowest midterm grade will be replaced by your grade on the final if you do better on the final. If you miss a midterm for any reason (illness, family emergency, didn't study, etc.), the final will be used to replace the midterm. Only one midterm grade can be replaced this way.
- Please notify me in writing by the second week of the term about any known or potential extracurricular conflicts with exam dates etc. (such as religious observances, graduate or medical school interviews, or team activities). I will try my best to help you with making accommodations but cannot promise them in all cases. In the event there is no mutually workable solution, you may be dropped from the class.

Grading policy:

Programming assignments: 8%
Homework assignments: 7%
Quizzes: 15%
Midterm 1: 20%
Midterm 2: 20%
Final exam: 30%

Grading scale:

Approximate final grade cutoffs:

A+: 96.7-100%	B+: 86.7-89.9%	C+: 76.7-79.9%	D+: 76.7-79.9%
A: 93.4-96.6%	B: 83.4-86.6%	C: 73.4-76.6%	D: 63.4-66.6%
A-: 90-93.3%	B-: 80-83.3%	C-: 70-73.3%	D-: 60-63.3%
			F: < 60%

Accommodations

UC Berkeley is committed to creating a learning environment that meets the needs of its diverse student body including students with disabilities. If you anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with me.
If you have a disability, or think you may have a disability, you can work with the Disabled Students' Program (DSP) to request an official accommodation. The Disabled Students' Program (DSP) is the campus office responsible for authorizing disability-related academic accommodations, in cooperation with the students themselves and their instructors. You can find more information about DSP, including contact information and the application process here: dsp.berkeley.edu.
Students who need academic accommodations or have questions about their accommodations should contact DSP. Students may call 642-0518 (voice), 642-6376 (TTY), or e-mail dsp@berkeley.edu.

Important things to note about accommodations:

- Accommodations are not retroactive. If you need accommodations, contact DSP early.

    - If you need DSP accommodations for an exam/quiz, we need to receive the official DSP accommodation letter from the DSP office at least 10 business days in advance. Given the expected large number of DSP students in the class, we will not be able to accommodate anyone in less than 10 business days. If late, the student will have to take the exam along with everyone else under the regular conditions provided for the class. The earlier we know of your official DSP status, the easier it is to provide appropriate accommodations. If you want to take advantage of any DSP accommodations granted beyond extended time for exams (e.g. distraction reduced environment etc.), contact me and your GSI at least one week before the event.

- Religious accommodations/student athletes: Discuss your schedule with instructor in office hours by the end of the second week of classes.

Questions about your DSP accommodations:

- re quizzes: speak to your GSI in advance, in office hours.

- re exams: speak to instructor in advance, in office hours.

Academic integrity

The University policy on academic integrity can be found at https://conduct.berkeley.edu/integrity/ . Any forms of cheating on quizzes or exams will be actively investigated and reported to ensure honesty and fairness in the class. It is the student's responsibility to read thoroughly the syllabus and be familiar with all the policies stated, in particular, the homework and exam policies. See also https://teaching.berkeley.edu/berkeley-honor-code for the Honor Code of UC Berkeley.

Wellness statement

Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, avoiding drugs and alcohol, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress.
All of us benefit from support during times of struggle. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is almost always helpful.
If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. Counseling and Psychological Services (CaPS) is here to help: call (855) 817-5667 and visit their website at https://uhs.berkeley.edu/caps. Consider reaching out to a friend, faculty or family member you trust for help getting connected to the support that can help. You can also check out this resource: https://recalibrate.berkeley.edu/.

If you have questions about this or your coursework, please let me know. Thank you, and have a great semester.